

SHYAM SUNDAR

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Professional Summary

Research Analyst specializing in academic scholar profiling, publication data extraction, and research impact analysis. Experienced in developing author-level dashboards and optimizing data workflows through scraping solutions and archival data retrieval strategies to improve record accuracy and operational efficiency

Work Experience

Research Analyst – Consultant

Aditya Softserv, Bengaluru

July 2024 – Present

- Scrape, validate, and structure academic publication data of an scholar's entire data management and updating for integration into Academic Analytics web platform
- Analysed publication networks, citation metrics, and authorship roles to improve author profiling accuracy and support data-driven research evaluation decisions
- Utilize Internet Archive (Wayback Machine) to recover missing scholarly records and improve data completeness
- Develop author-level dashboards using proprietary data visualization tools to support data-driven decision-making
- Proposed and implemented workflow optimization strategies that significantly reduced manual data correction efforts

Internship Experience

Project Intern

ICAR– Sugarcane Breeding Institute, Coimbatore, IN

June 2023 – July 2023

- Performed PCR-based molecular characterization of 20+ sugarcane clones, enabling genotype differentiation for breeding evaluation studies
- Performed Agarose Gel Electrophoresis (AGE) and Polyacrylamide Gel Electrophoresis (PAGE) for DNA fragment analysis
- Assisted in meristem harvesting for clonal propagation and genetic evaluation studies
- Maintained laboratory documentation and ensured adherence to molecular biology protocols

Production Intern

Annai Agri Tech, Sivagangai, IN

December 2022

- Contributed to the production and formulation of biofertilizers and bio stimulants, ensuring adherence to microbial concentration standards across production batches
- Executed quality control analysis on final products to ensure compliance with microbial count and stability standards
- Monitored fermentation parameters and maintained production records to ensure process consistency
- Supported laboratory testing and validation procedures for product performance

Skills

Programming & Data Handling

- Python – Pandas, NumPy, scikit-learn
- Applied Machine Learning using scikit-learn (Supervised & Unsupervised)
- Model Evaluation & Validation

Analytics & Visualization

- Data Driven Decision making
- Research Impact Analysis
- Data Validation & Quality Assurance

Statistical & Optimization Tools

- Design-Expert (DoE & RSM)
- Process Optimization
- Statistical Modeling
- Experimental Design

Education

Bachelor of Technology (B.Tech.) in Biotechnology

Sethu Institute of Technology, Virudhunagar

2020 – 2024

CGPA: 8.5 / 10

Certifications

Master Program on Data Science & Artificial Intelligence

FITA Academy | 2026

Key Topics: Python, Machine Learning, Data Visualization, Statistical Analysis

Bioinformatics with Python

Udemy | 2023

Focus: Sequence data handling, biological data parsing

Fundamentals of Digital Marketing

Google Digital Garage | 2023

Technical Projects

Academic Author Database using OpenAlex API (Ongoing)

Tools: Python, OpenAlex API, JSON Parsing, Data Structuring

- Built an automated Python-based pipeline integrating OpenAlex API to extract and structure entire publication records of academic scholars for scalable research.
- Parsed and structured author-level data including publications, citation counts, affiliations, and research domains
- Implemented data cleaning and normalization procedures to ensure consistency and accuracy

R-JD-Analyser-Cloud

Tools: Python, Ollama, JSON Parsing, Data Structuring, Langchain, Streamlit

GitHub: github.com/Shyam-2003/R-JD-Analyser-Cloud

- Created a cloud-based job description analysis tool using python
- Processed and analysed 200+ job descriptions using LLM Based Processing and keyword extraction to identify high-demand technical skill clusters
- Analysed job descriptions to identify skill frequency and domain-specific requirements
- Generated structured insights to assist in resume alignment and skill gap analysis

Fabrication of Smart Orthotic Callipers with Nano Fabrics for Physically Challenged Persons

MSME-Funded Academic Project

- Initiated to build lightweight orthotic callipers using nano-fabric composites to improve durability, comfort, and mobility support.
- Conducted material characterization and biomechanical suitability testing.
- Designed with focus on cost-effectiveness and accessibility for physically challenged individuals.
- Implementing project under financial support from the Ministry of Micro, Small and Medium Enterprises (MSME), Government of India.

Production and Optimization of Herbal Soap

Tools: Design-Expert (DoE, RSM)

- Applied Response Surface Methodology (RSM) using Design-Expert to optimize formulation parameters, improving product stability and reducing material variability
- Optimized formulation parameters to improve product stability and TFM Total fatty matter is shows were 76.884 % (Grade I)
- Performed statistical validation using ANOVA
- Identified optimal parameter combinations through model prediction

Phytochemical Characterization of Linum Usitatissimum & Sesamum Indicum

- Performed extraction and phytochemical screening
 - Performed comparative analysis of bioactive compounds
 - Optimized treatment parameters for ulcerative colitis application and Anti-Inflammatory percentage were 75.4%
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